CPE 334 | Lab 4

TokTickIT

Actions Taken, dashboards, and final regression

Release evidence and demonstration report

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Peer reviewer	MadMax168 (GitHub)
Repository	ovenmakemeheat/toktickit
Parent issue	#86 – Lab 4
Report branch	`docs/lab4-docs`
Release source	`main`, `bd8cecd` (PR #98)
Stack	React, TypeScript, Vite, Express, Prisma, PostgreSQL

Release source: PR #98 merged `lab4-staging` into `main` on 2026-10-29 at `bd8cecd63f1dd3386b0b7f2ffcf270805e0190f3`. The report is prepared on `docs/lab4-docs` from the release source tree.

This PDF uses the required Answer Part 1 through Answer Part 9 headings in order.

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Answer Part 1: Git Use with Engineering Workflow

Lab 4 used one feature branch per approved issue, integrating feature Pull Requests into `lab4-staging` before the release Pull Request to `main`. The table lists linked branches, associated Pull Requests, and merge destinations.

Issue	Feature branch / PR	Integration
#87 Contract	`feature/20-lab4-contract-test-plan`; PR #91	Merged to `lab4-staging` at `c945062` before implementation PRs.
#88 Actions/workflow	`feature/21-lab4-actions-taken-ticket-workflow`; PR #92	Merged to `lab4-staging` at `8608a31`.
#89 Dashboards	`feature/22-lab4-role-dashboards`; PR #93	Merged to `lab4-staging` at `47d6487`.
#90 Regression/readiness	`feature/23-lab4-regression-demo-readiness`; PR #94	PR #94 merged to staging at `7cb1715`; follow-ups #95, #96, #97 also merged to staging.
Release	`lab4-staging`; PR #98	Merged by MadMax168 into `main` on 2026-10-29 at `bd8cecd`.

The release commit is `bd8cecd`. At release, `main` and `lab4-staging` had identical Git trees. As captured on 2026-10-29, all Lab 4 issues #86–#90 are closed; parent issue #86 is in `Done` on the public Lab 4 Project Kanban, with native sub-issue progress at 4/4 (100%). The board screenshot below records the visible final state. Native sub-issues remain attached to the parent and are not separate board items.

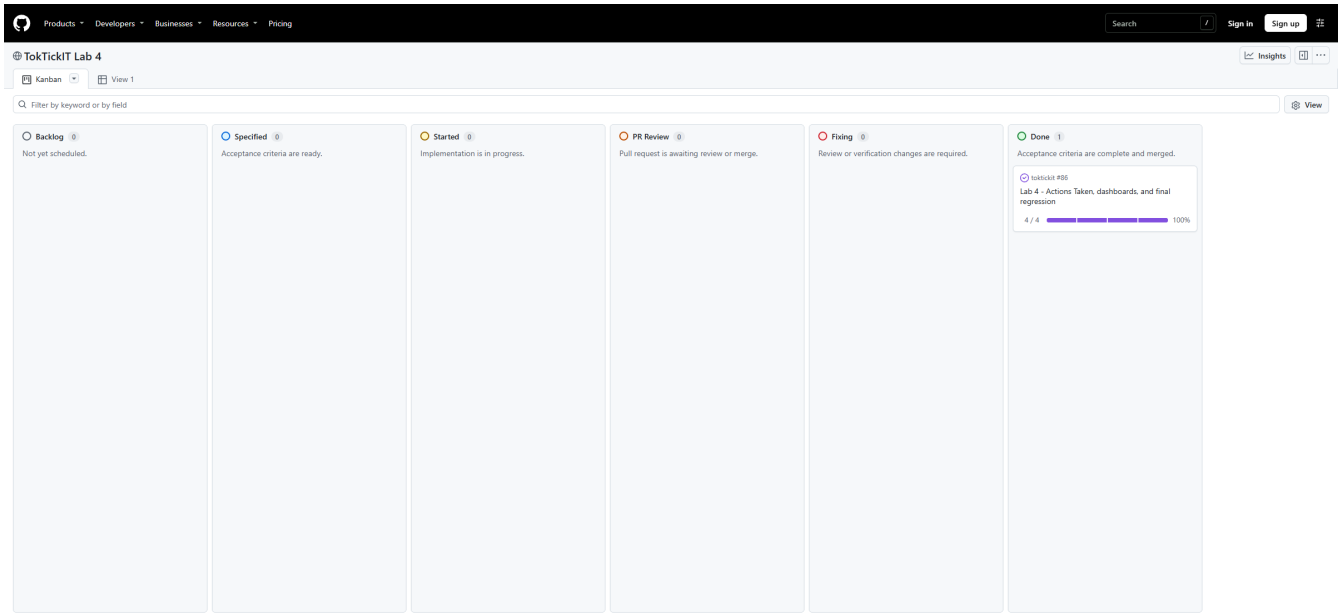


Figure 1: Lab 4 Project Kanban captured 2026-10-29: parent issue #86 is in Done and reports 4/4 sub-issues complete.

Reviewer and repository evidence

The feature and release Pull Requests are linked in the table above; MadMax168 is identified as peer reviewer, and PR #98 records the release merge to `main`. The engineering contract PR #91 predates implementation PRs #92–#94. The report branch is based on the released source tree and contains this documentation work.

> **Repository structure at release:** `client/` (React application and UI tests); `server/` (Express API, Prisma schema/migrations, and API tests); `e2e/` (Playwright journeys); `docs/lab-04/` (engineering contract, test plan, evidence, and report); `artifacts/lab-04/` (visual and machine-readable release evidence).

The linked README documents Bun/Docker setup, migration, seed, test database, local demonstration, and verification commands. The linked `.gitignore` excludes environment secrets, local database/storage state, build and test output, and LaTeX intermediates.

Answer Part 2: Spec DD

The Lab 4 engineering contract converts the handout into a bounded, testable increment. The contract existed and was integrated through PR #91 before the dependent implementation PRs. It covers **FR-01**–**FR-20**, **BR-01**–**BR-25**, and **AC-01**–**AC-26**; the complete rendered source is `specification.md`, with supporting API, UI, and test contracts.

- **Actions Taken:** child of exactly one Ticket; actor derived from the authenticated session; action time and stable chronological order; required description/result; conditional follow-up note; optional plain-text Attachment Notes; idempotent creation; versioned edits; no deletion.
- **Workflow:** only IT Staff/Admin may transition status; required confirmations are explicit; 'RESOLVED' requires real work, and re-resolution after reopening requires a newly created Action Taken. Requester resolution indication remains advisory.
- **Dashboards:** requester-owned metrics and staff/admin operational metrics are computed server-side using one 'asOf' and an inclusive rolling 30-day UTC window. Recent lists are bounded to five rows and drill into existing detailed screens.
- **Preservation:** additive migration retains Lab 3 rows and relationships, fabricates no work history, and documents timestamp approximations for legacy resolved/closed/reopened Tickets. Seed behavior is repeat-safe.

Two consequential database choices are an immutable 'createdAt' boundary for the post-reopen work gate (because editable 'actionAt' does not establish when a record was added) and a separate 'resolvedAt' value (because comments/actions can update Ticket activity without changing resolution time). The migration adds 'lastReopenedAt' as an internal marker and avoids a destructive rollback after new work records exist.

Numbered requirements and acceptance coverage

ID	Contracted behavior
FR-01	IT Staff/Admin create Actions Taken; performer comes from the authenticated session.
FR-02	Store action time, description, result, follow-up flag/note, and Attachment Notes.
FR-03	Requesters read work only on owned Tickets; no Action Taken writes.
FR-04	Staff/Admin edits preserve original performer, Ticket, and creation data.
FR-05	Stable ordering, idempotent create, and stale-edit conflicts.
FR-06	No Action Taken delete operation.
FR-07	Server-enforced status matrix and confirmations.
FR-08	Resolution requires real work; Requester signal is advisory.
FR-09	Track resolution and reopen timestamps with documented legacy backfill.
FR-10	Requester dashboard contains only authenticated Requester metrics/summaries.
FR-11	Staff/Admin dashboard uses owner, status, priority, and current-user filters.
FR-12	Dashboards return bounded summaries and actionable drill-downs.
FR-13	Dashboard calculations use a single server-side UTC window.
FR-14	Additive migration preserves Lab 3 data; no fabricated Actions Taken.
FR-15	Repeat-safe seed covers representative statuses and zero/non-zero data.
FR-16	Lab 1–3 authentication, role, Ticket, attachment, comment, note, and admin behavior remains.
FR-17	Role screens remain keyboard/assistive-technology usable and responsive.
FR-18	Acceptance evidence maps to tests, commands, and manual checks.
FR-19	Staff/Admin 'myRecentActions' is performer-scoped, status-independent, and bounded.
FR-20	Release hardening checks role navigation, page errors, links, and unfinished UI.

Rule / criteria	Summary and test trace
BR-01–BR-15	Ticket-owned Action Taken data, immutable actor/parent/fingerprint, editable audit fields, follow-up rules, stable order, idempotency, versions, role/ownership enforcement, and atomic activity updates. Verified by UNIT-01, API-01–04, E2E-01.
BR-16–BR-24	Complete transition matrix; confirmation; Action Taken resolution gate; 'resolvedAt'/'lastReopenedAt'; atomic status conflict; legacy Tickets stay valid. Verified by UNIT-02, API-05/API-08, E2E-02.
BR-25; AC-25	Current-user 'My Recent Actions' is performer-scoped rather than owner-scoped, includes every Ticket status, observes inclusive 30-day UTC bounds, sorts newest-first, and caps at five. Verified by API-07, UI-02, E2E-03, PERF-01.
AC-01–AC-09	Action Taken create/read/update, role boundaries, safe attachment notes, idempotency, and stale-version behavior. Verified by UNIT-01, API-01–04, UI-01, E2E-01.
AC-10–AC-12	Status matrix, resolution/reopen gates, confirmation, and concurrent status conflict. Verified by UNIT-02, API-05, E2E-02.
AC-13–AC-17	Requester and operational dashboard query scopes, calculations, bounded rows, destinations, and Admin capability retention. Verified by API-04/API-06/API-07, UI-02/UI-03, E2E-03/E2E-04.
AC-18–AC-24	UI states, Labs 1–3 regression, migration/seed, responsive/accessibility, performance, and contract evidence. Verified by UI-01–03, API-08, PERF-01, VIS-01, DOC-01.
AC-26	Representative seeded release routes have role/API evidence, zero unexpected page/console errors, working navigation, and no visible placeholder text. Verified by E2E-04 and VIS-01.

Product Definition of Done: the implemented features, migration/recovery tests, seeded regression suite, responsive/accessibility inspection, prior-Lab regression, and setup/demo documentation are recorded complete. At report capture, issues #86–#90 were closed and the Kanban placed parent #86 in Done with all four native sub-issues complete; see the board evidence above.

Answer Part 3: Test DD and Traceability

The acceptance criteria map to named unit, API/integration, UI, E2E, performance, visual, and documentation checks in `tests.md`. Representative files include:

Coverage	Actual test files
Actions Taken validation, persistence, actor/owner, idempotency, edit conflicts	`server/tests/lab-04/actions-taken.unit.test.ts`; `server/tests/lab-04/actions-taken.api.test.ts`; `client/tests/lab-04/ActionsTaken.test.tsx`
Ticket transition/resolution rules and role access	`server/tests/lab-04/ticket-workflow.api.test.ts`; `client/tests/lab-04/TicketWorkflow.test.tsx`
Requester and operational dashboard metrics	`server/tests/lab-04/requester-dashboard.api.test.ts`; `server/tests/lab-04/staff-dashboard.api.test.ts`; `client/tests/lab-04/RequesterDashboard.test.tsx`; `client/tests/lab-04/StaffDashboard.test.tsx`
Migration, seed recovery, and performance smoke	`server/tests/lab-04/migration-seed.api.test.ts`; `server/tests/lab-04/dashboard-performance.api.test.ts`
Real seeded E2E and prior-Lab regression	`e2e/lab-04/actions-taken-flow.spec.ts`; `e2e/lab-04/ticket-resolution.spec.ts`; `e2e/lab-04/dashboards.spec.ts`; `e2e/lab-04/prior-labs-regression.spec.ts`; Lab 2/3 release suites

Executed against release source `bd8cecd63f1dd3386b0b7f2ffcf270805e0190f3`, from the report branch with the same product tree as `main`. 'bun run verify' passed: Biome checked 142 files; Lefthook and Prisma validation passed; client/server type checks and builds passed; client tests passed (19 files/72 tests); server tests passed (29 files/171 tests). 'bun run test:e2e' passed all 19 Chromium tests across Labs 2–4 against isolated PostgreSQL `toktickit_test` on port 15434. Seeded Lab 4 journeys used real APIs without business-route mocks. E2E-04 records the release revision, zero browser/page errors, and 'businessApiMocks: false'.

Complete captured command output: the logs below are sanitized terminal output from the passing commands. ANSI color/control sequences and non-ASCII terminal decoration are normalized for PDF rendering; test names, summaries, and command results are retained.

`bun run verify`

```
$ bun run check && bun run hooks:validate && bun run db:validate && bun run typecheck && bun run test && bun run build
$ biome check .
Checked 142 files in 220ms. No fixes applied.
$ lefthook validate
All good
$ bun run --cwd server prisma:validate
$ prisma validate
Loaded Prisma config from prisma.config.ts.

Prisma config detected, skipping environment variable loading.
Prisma schema loaded from prisma\schema.prisma
The schema at prisma\schema.prisma is valid
$ bun run typecheck:client && bun run typecheck:server
$ bun run --cwd client typecheck
$ tsc -b --pretty false
$ bun run --cwd server typecheck
$ tsc -p tsconfig.json --noEmit
$ bun run test:client && bun run test:server
$ bun run --cwd client test
$ vitest run

 RUN  v3.2.7 C:/Users/Gunte/Workspace/Workspace/works/CPE/toktickit/client

 PASS tests/lab-03/StaffTicketQueue.test.tsx (4 tests) 1031ms
   PASS IT Staff Ticket Queue > renders query controls, operational fields, and opens detail  630ms
 PASS tests/lab-03/AdminTicketReview.test.tsx (3 tests) 1112ms
   PASS Administrator Ticket Review > shows read-only Ticket facts and only permits IT Priority changes  856ms
 PASS tests/lab-04/RequesterDashboard.test.tsx (3 tests) 948ms
   PASS Lab 4 Requester Dashboard > shows scoped summary cards and recent Ticket details  530ms
 PASS tests/lab-04/StaffDashboard.test.tsx (4 tests) 1067ms
   PASS Lab 4 Staff Dashboard > shows active metrics and opens the Queue with its dashboard filters intact  707ms
 PASS tests/lab-03/ApplicationShell.test.tsx (5 tests) 1105ms
   PASS Lab 3 authenticated application shell > shows the IT_STAFF identity without requester controls  464ms
 PASS tests/lab-02/AttachmentSection.test.tsx (3 tests) 1615ms
   PASS Issue #55 AttachmentSection > rejects an invalid file before calling the upload API  510ms
   PASS Issue #55 AttachmentSection > uploads a permitted file with session ownership and offers retry on failure  380ms
   PASS Issue #55 AttachmentSection > downloads active files and requires confirmation plus a reason before removal  724ms
 PASS tests/lab-03/StaffTicketDetail.test.tsx (3 tests) 1702ms
   PASS IT Staff Ticket Detail > shows read-only facts, private/public communication, attachments, and operations  536ms
   PASS IT Staff Ticket Detail > shows refresh feedback for a concurrent status conflict  560ms
```

```
   PASS IT Staff Ticket Detail > requires status confirmation and submits permitted mutations  604ms
 PASS tests/lab-02/RequesterTicketDetail.test.tsx (4 tests) 1796ms
   PASS Issue #55 Requester Ticket Detail > renders owned ticket fields read-only and separates active and removed attachments  609ms
   PASS Issue #55 Requester Ticket Detail > keeps upload success feedback visible while refreshing the Ticket  313ms
   PASS Issue #55 Requester Ticket Detail > lets the Requester post a Public Comment and indicate resolution without closing  811ms
 PASS tests/lab-02/MyTickets.test.tsx (5 tests) 2036ms
   PASS Lab 3 authenticated My Tickets > shows loading, authenticated results, and the documented ticket fields  805ms
   PASS Lab 3 authenticated My Tickets > distinguishes an empty requester list from a no-results query and clears filters  431ms
   PASS Lab 3 authenticated My Tickets > opens a ticket detail view from the result action  307ms
 PASS tests/lab-03/ChangePassword.test.tsx (3 tests) 2110ms
   PASS Lab 3 mandatory initial password change > blocks the normal shell and explains the password rules  303ms
   PASS Lab 3 mandatory initial password change > shows client validation and does not submit an invalid password  969ms
   PASS Lab 3 mandatory initial password change > saves a valid password and then opens the authenticated shell  837ms
 PASS tests/lab-03/Login.test.tsx (5 tests) 2420ms
   PASS Lab 3 login > starts with a focused sign-in form and no requester selector  341ms
   PASS Lab 3 login > shows a generic invalid-credential failure and clears the password  702ms
   PASS Lab 3 login > opens the authenticated shell after a successful login  835ms
   PASS Lab 3 login > explains an inactive account without exposing credential details  370ms
 PASS tests/lab-02/ZenGreenResponsive.test.tsx (4 tests) 2952ms
   PASS Lab 3 Zen Green requester regression > exposes the approved tokens and accessible shared control states  810ms
   PASS Lab 3 Zen Green requester regression > exposes field-level invalid state through accessible relationships  500ms
   PASS Lab 3 Zen Green requester regression > marks the current requester page in the application navigation  411ms
   PASS Lab 3 Zen Green requester regression > exposes busy and success states without allowing duplicate submission  1229ms
 PASS tests/lab-04/ActionsTaken.test.tsx (5 tests) 3512ms
   PASS Actions Taken panel > records required work details, follow-up, and plain-text attachment notes  2539ms
   PASS Actions Taken panel > edits mutable fields with optimistic version and preserves original performer  380ms
   PASS Actions Taken panel > refreshes the current version after conflict without discarding the draft  317ms
 PASS tests/lab-02/attachment-policy.test.ts (4 tests) 10ms
 PASS tests/lab-01/App.test.tsx (1 test) 192ms
 PASS tests/lab-02/RequesterSelection.test.tsx (2 tests) 298ms
 PASS tests/lab-03/UserManagement.test.tsx (6 tests) 4555ms
   PASS Administrator User Management > lists the required columns with search, role filter, and actions  1154ms
   PASS Administrator User Management > creates a User with the required identity and initial password fields  1392ms
   PASS Administrator User Management > blocks weak passwords and surfaces duplicate-email conflicts  980ms
   PASS Administrator User Management > edits an account, prevents self-deactivation, and sets a new initial password  866ms
 PASS tests/lab-04/TicketWorkflow.test.tsx (2 tests) 737ms
   PASS Administrator Ticket operations > allows operational work and Actions Taken while keeping communications read-only  549ms
 PASS tests/lab-02/CreateTicket.test.tsx (6 tests) 5236ms
   PASS Lab 3 authenticated Create Ticket regression > shows a loading state before active reference controls are ready  569ms
   PASS Lab 3 authenticated Create Ticket regression > submits one valid ticket with authenticated ownership  1681ms
   PASS Lab 3 authenticated Create Ticket regression > disables Submit and prevents duplicate requests while busy  992ms
   PASS Lab 3 authenticated Create Ticket regression > preserves entered values after a safe API failure  846ms
   PASS Lab 3 authenticated Create Ticket regression > rejects unsupported attachments before calling the create API  958ms

 Test Files  19 passed (19)
      Tests  72 passed (72)
   Start at  13:56:37
   Duration  15.13s (transform 3.81s, setup 33.42s, collect 20.76s, tests 34.43s, environment 77.26s, prepare 7.92s)

$ bun run --cwd server test
$ vitest run

 RUN  v4.1.10 C:/Users/Gunte/Workspace/Workspace/works/CPE/toktickit/server


 Test Files  29 passed (29)
      Tests  171 passed (171)
   Start at  13:56:55
   Duration  82.39s (transform 1.12s, setup 652ms, import 21.04s, tests 50.88s, environment 4ms)

$ bun run build:client && bun run build:server
$ bun run --cwd client build
$ tsc -b && vite build
vite v7.3.6 building client environment for production...
transforming...
PASS 49 modules transformed.
rendering chunks...
computing gzip size...
dist/index.html                   0.49 kB | gzip:  0.30 kB
dist/assets/index-DfiwSy_-.css  250.27 kB | gzip: 34.44 kB
dist/assets/index-D1GCJ51X.js   333.76 kB | gzip: 88.43 kB
PASS built in 1.64s
$ bun run --cwd server build
$ tsc -p tsconfig.json
```

bun run test:e2e

```
$ bun run db:test:setup && playwright test
$ bun run --cwd server db:test:setup
$ tsx scripts/prepare-test-db.ts
Loaded Prisma config from prisma.config.ts.

Prisma config detected, skipping environment variable loading.
Prisma schema loaded from prisma\schema.prisma
Datasource "db": PostgreSQL database "toktickit_test", schema "public" at "localhost:15434"

Applying migration `20260805160550_add_category`
Applying migration `20260824091500_add_lab2_requester_context`
Applying migration `20260913073929_lab3_authentication`
Applying migration `20260914140000_enforce_requester_ticket_ownership`
Applying migration `20261008090000_lab4_actions_taken_workflow`
Applying migration `20261011120000_lab4_dashboard_indexes`

Database reset successful

The following migration(s) have been applied:

migrations/
  +- 20260805160550_add_category/
    +- migration.sql
  +- 20260824091500_add_lab2_requester_context/
    +- migration.sql
  +- 20260913073929_lab3_authentication/
    +- migration.sql
  +- 20260914140000_enforce_requester_ticket_ownership/
    +- migration.sql
  +- 20261008090000_lab4_actions_taken_workflow/
    +- migration.sql
  +- 20261011120000_lab4_dashboard_indexes/
    +- migration.sql

Running generate... (Use --skip-generate to skip the generators)
Running generate... - Prisma Client
PASS Generated Prisma Client (v6.19.0) to .\..\node_modules\.bun\@prisma+client@6.
19.0+62a78c54b6dcb451\node_modules\@prisma\client in 170ms

Loaded Prisma config from prisma.config.ts.

Prisma config detected, skipping environment variable loading.
Running seed command `tsx prisma/seed.ts` ...
TokTickIT Lab 3 seed completed.

The seed command has been executed.
[WebServer] $ bun run --cwd server dev
[WebServer] $ tsx watch src/index.ts
[WebServer] $ bun run --cwd client dev
[WebServer] $ vite --host 0.0.0.0

Running 19 tests using 1 worker

  ok  1 [chromium] > e2e\lab-02\report-evidence.spec.ts:132:7 > Issue #65 report evidence > captures authenticated requester, create, and query-control states (4.1s)
  ok  2 [chromium] > e2e\lab-02\requester-ticket-flow.spec.ts:180:7 > Authenticated Requester Lab 2 regression > completes the authenticated ticket and attachment lifecycle (15.6s)
  ok  3 [chromium] > e2e\lab-03\authentication-and-requester.spec.ts:141:7 > Lab 3 authentication and Requester regression > protects direct navigation, supports login, and invalidates logout
       (1.5s)
  ok  4 [chromium] > e2e\lab-03\authentication-and-requester.spec.ts:177:7 > Lab 3 authentication and Requester regression > explains invalid credentials and inactive accounts safely (1.0s)
  ok  5 [chromium] > e2e\lab-03\authentication-and-requester.spec.ts:233:7 > Lab 3 authentication and Requester regression > requires an initial password change before showing normal
      navigation (965ms)
  ok  6 [chromium] > e2e\lab-03\authentication-and-requester.spec.ts:267:7 > Lab 3 authentication and Requester regression > keeps the Lab 2 Requester flow under the authenticated identity
       (1.7s)
  ok  7 [chromium] > e2e\lab-03\release-regression.spec.ts:80:7 > Lab 3 integrated release regression > runs the seeded API through the role route smoke matrix (2.3s)
  ok  8 [chromium] > e2e\lab-03\release-regression.spec.ts:134:7 > Lab 3 integrated release regression > enforces role-forbidden API and destination boundaries with a seeded session (1.1s)
  ok  9 [chromium] > e2e\lab-03\responsive-and-accessibility.spec.ts:196:7 > Lab 3 responsive and accessible presentation > switches the Staff Queue between cards and table without overflow
       (863ms)
  ok 10 [chromium] > e2e\lab-03\responsive-and-accessibility.spec.ts:239:7 > Lab 3 responsive and accessible presentation > captures the Requester routes at every required viewport (5.4s)
  ok 11 [chromium] > e2e\lab-03\responsive-and-accessibility.spec.ts:275:7 > Lab 3 responsive and accessible presentation > keeps Administrator User Management cards and named controls
      accessible on tablet (1.1s)
  ok 12 [chromium] > e2e\lab-03\staff-ticket-flow.spec.ts:237:7 > Lab 3 IT Staff Ticket workflow > completes queue, ownership, priority, status, and communication operations (4.2s)
  ok 13 [chromium] > e2e\lab-03\user-administration.spec.ts:233:7 > Lab 3 Administrator management > manages Users and performs read-only Ticket Review with IT Priority access (4.1s)
  ok 14 [chromium] > e2e\lab-03\user-administration.spec.ts:343:7 > Lab 3 Administrator management > does not expose Administrator screens to IT Staff (547ms)
  ok 15 [chromium] > e2e\lab-04\actions-taken-flow.spec.ts:18:7 > Lab 4 seeded Actions Taken journey > Staff creates and edits work; the owning Requester sees read-only history (4.0s)
  ok 16 [chromium] > e2e\lab-04\dashboards.spec.ts:154:7 > Lab 4 role dashboards > shows session-scoped Requester and performer-scoped Staff/Admin dashboards (3.3s)
  ok 17 [chromium] > e2e\lab-04\prior-labs-regression.spec.ts:247:7 > Lab 4 integrated role and prior-Lab release regression > enforces the seeded role matrix and audits representative routes
       (6.6s)
  ok 18 [chromium] > e2e\lab-04\ticket-resolution.spec.ts:48:7 > Lab 4 seeded resolution gate journey > keeps the Requester signal advisory and resolves only after work and confirmation (4.3s)
  ok 19 [chromium] > e2e\lab-04\ticket-resolution.spec.ts:177:7 > Lab 4 seeded resolution gate journey > requires new work recorded after reopening before resolving again (4.2s)

  19 passed (1.3m)
```

Answer Part 4: AI Use with Reflection

The specification and Issue #90 implementation sessions used OpenAI Codex.

> Model: `gpt-6-luna`
> Provider: `openai-codex`

The full AI-use record includes the human-responsibility statement. Nine selected prompts are:

1. “on @docs/lab-04/requirements/UTF-8_SE+Lab+4.pdf”
2. “this should break down into tickets issues (use my issues format)”
3. “make sure you keep every requirement”
4. “create a project board for lab 4”
5. “using kanban as main”
6. “next let’s create lab 4 staging branch”
7. “edit issue propose branch number to increment continuing with lab3 number branch name”
8. “now get into first branch and start implement”
9. “Implement Issue #90.”

My Reflection. I used the specification agent to turn the Lab 4 handout into numbered requirements, business rules, acceptance criteria, and a test plan before implementation. The coding agent helped implement the approved work and prepare seeded regression journeys and release evidence. I reviewed the scope against the handout, inspected the changes, and used the verification and end-to-end results to assess the released product. A key technical lesson from the work was that ‘updatedAt‘ cannot prove new work was recorded after a Ticket is reopened: ordinary edits also change activity timestamps. The separate ‘lastReopenedAt‘ marker and Action Taken creation time make that gate testable. AI accelerated contract drafting and repetitive implementation, while I remained accountable for scope, evidence, and the final decisions.

Answer Part 5: Working IT Staff Dashboard UI

IT Staff and Administrators share the operational Dashboard. It reports active unassigned work, the signed-in User's owned active work, active high priority work, status and IT Priority breakdowns, recently updated Tickets, and the current User's recent Actions Taken. Calculations are server-backed, bounded, and linked to Queue/Ticket Detail destinations; the Administrator reuses the IT Staff view.

The seeded dashboard inspection recorded 18 active Tickets: 14 New, 1 Open, 1 In Progress, 1 Waiting for Requester, and 1 Reopened. IT Priority totals were 1 Low, 1 Medium, and 16 High. The same sample included 14 unassigned active Tickets, 0 assigned to the inspected IT Staff account, and 16 high-priority active Tickets. API tests compare exact query results, UTC-window boundaries, scopes, caps, and safe DTOs against deterministic database fixtures. The local populated-seed performance smoke recorded Staff 20.5 ms / 3,167 B and Requester 14.1 ms / 2,272 B; these are local measurements, not production SLOs. Loading, empty, forbidden, retryable failure, and drill-down states are covered by the client/API tests and seeded E2E suite.

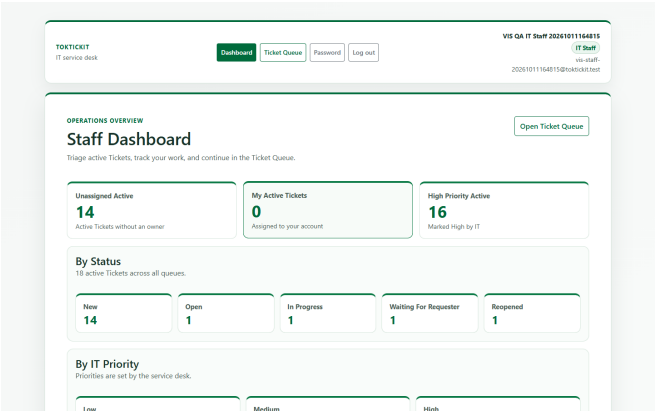


Figure 2: Seeded Staff Dashboard metrics and breakdowns.

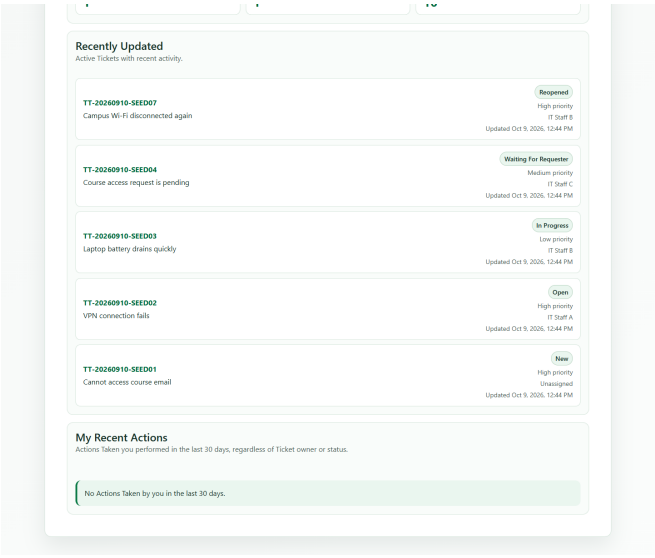


Figure 3: Recently updated Tickets and the empty current-user Actions Taken state.

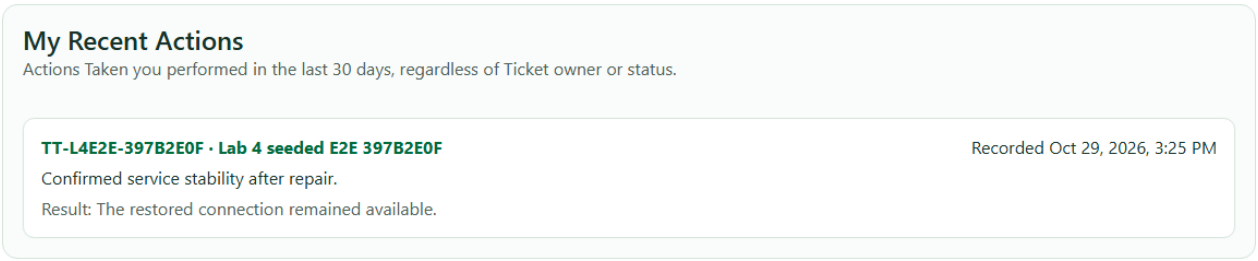


Figure 4: A seeded Action Taken performed by IT Staff A appears in My Recent Actions even though IT Staff B owns the resolved Ticket. Tablet and mobile component captures are also recorded.

Answer Part 6: Working Actions Taken UI

The staff Ticket Detail presents work history separately from public communication, private Internal Notes, and Attachments. Each Action Taken shows the authenticated performer, action time, description, result, and follow-up state. Staff/Admin may add or edit records; Requesters see owned-ticket history read-only. The API keeps the primary Ticket Owner unchanged, rejects inactive assignment targets, prevents retry duplicates, and preserves the winning edit on stale-version conflicts. There is no delete or upload control in the Action Taken section.

The seeded E2E journey creates and edits Actions Taken under real PostgreSQL, verifies Requester read-only visibility and write denial, and checks that repeated creation does not duplicate work. The visual evidence below shows two chronological work records on the same Ticket. Client/API tests cover validation, safe failures, role restrictions, follow-up note rules, plain-text Attachment Notes, and optimistic edits. Ticket Owner assignment, status changes, resolve/cancel paths, and inactive-assignee rejection are exercised by the Ticket workflow and release-regression suites; the Action Taken performer remains distinct from the Ticket Owner.

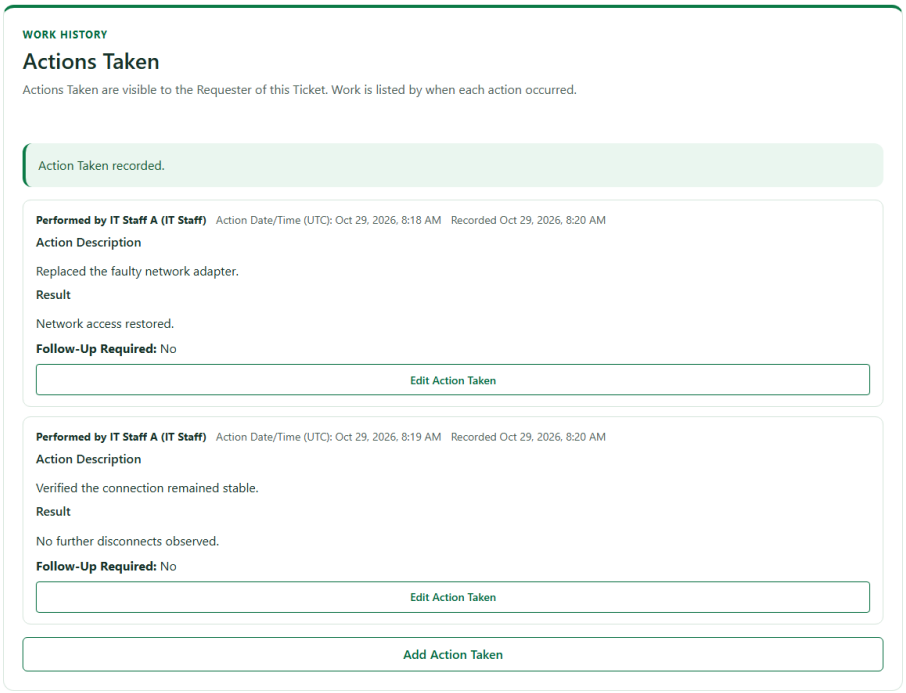


Figure 5: Two Actions Taken on one seeded Ticket in the Staff view. The Staff can edit each entry; creation time and action time remain distinct.

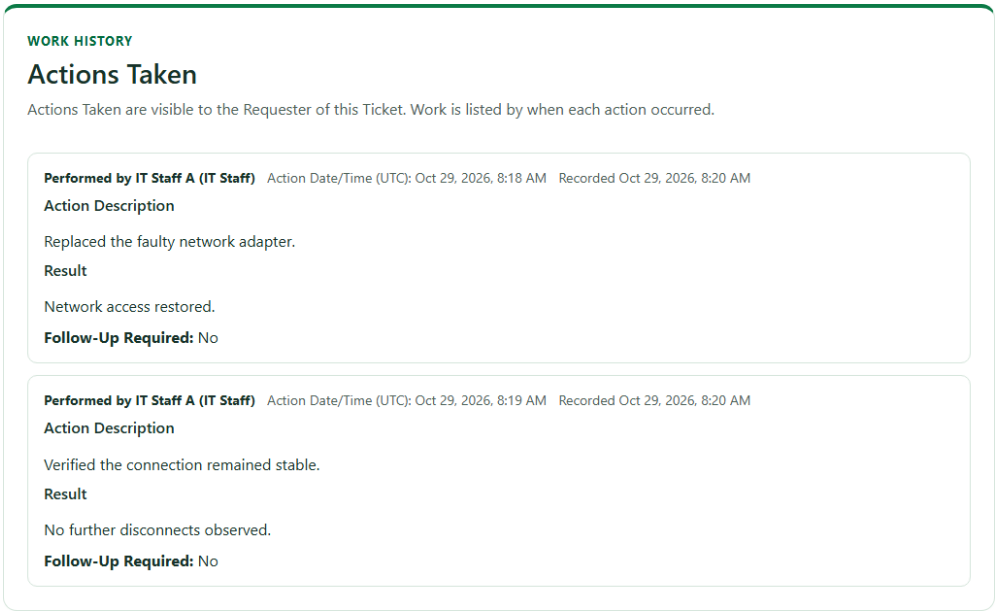


Figure 6: The same two Actions Taken are visible to the owning Requester; create and edit controls are absent. The test used the real API and isolated PostgreSQL. Create/edit forms at all viewports are included in Part 9.

Answer Part 7: Working Ticket Workflow

Only IT Staff and Administrators may perform formal transitions. The backend rejects unlisted transitions and requires explicit confirmation for resolution, closure, reopening, and cancellation. New resolution requires an Action Taken; after reopening, the work record must have been created after the latest reopen. Requester indication remains advisory. Action history orders by ‘actionAt‘, then ID; entries are not deletable, and edits preserve original performer/creation metadata while incrementing a version.

Current status	Permitted next status
NEW	OPEN; CANCELLED (confirmation)
OPEN	IN_PROGRESS; WAITING_FOR_REQUESTER; RESOLVED (confirmation and work); CANCELLED (confirmation)
IN_PROGRESS	OPEN; WAITING_FOR_REQUESTER; RESOLVED (confirmation and work); CANCELLED (confirmation)
Waiting for Requester	OPEN; IN_PROGRESS; RESOLVED (confirmation and work); CANCELLED (confirmation)
RESOLVED	CLOSED (confirmation); REOPENED (confirmation)
CLOSED	REOPENED (confirmation)
REOPENED	OPEN; IN_PROGRESS; WAITING_FOR_REQUESTER; RESOLVED (confirmation and new work); CANCELLED (confirmation)
CANCELLED	REOPENED (confirmation)

API and unit tests cover the full transition matrix, confirmation, the resolution gate, reopen marker, status conflicts, Requester advisory behavior, stable Action Taken ordering, and role visibility. The Ticket Detail screenshot in Part 6 also shows the workflow controls and status separately from work history.

Answer Part 8: Working Requester Dashboard and Final Regression UI

The Requester Dashboard scopes every count, recent Ticket, and drill-down to the authenticated Requester. It presents open, waiting, recently updated, and recently resolved work without replacing My Tickets. A seeded zero-ticket account shows explicit zero counts and empty states. Direct cross-owner requests return non-disclosing results, and Requesters cannot write Actions Taken or perform staff status transitions.

The complete seeded release suite also preserves Lab 1–3 login/password behavior, Requester ticket creation and ownership, Attachments, Public Comments, IT Staff queue/status work, Internal Notes, Administrator User Management, and Ticket Review. See the E2E-04 route/role/API report and the corresponding Lab 2/3 release test suites.

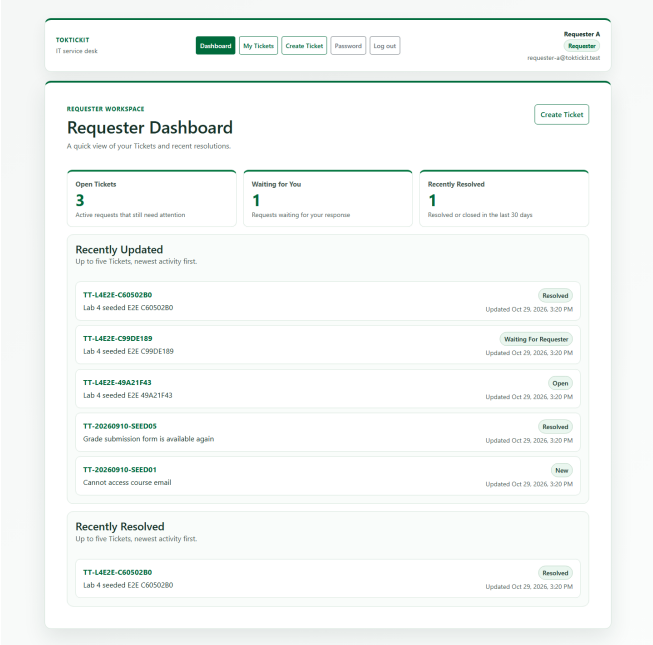


Figure 7: Seeded Requester dashboard with owned active, waiting, recently updated, and recently resolved Tickets.

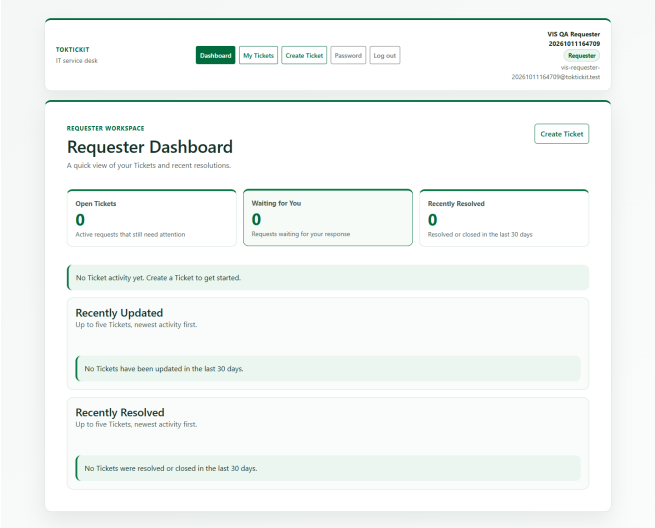


Figure 8: Zero-data Requester dashboard with zero counts and explicit empty states.

Counts and rows in the populated state are produced by the real API from isolated PostgreSQL fixtures; the empty state uses a Requester with no Tickets.

Answer Part 9: Zen Green UI, Responsive, Accessibility, and Final Polish

The UI preserves the Lab 3 Zen Green tokens and shared navigation. The rendered summary below and the linked full UI specification cover role navigation, dashboard cards, Actions Taken, loading/validation/empty/error feedback, and responsive behavior. VIS-01 records Requester, Staff, and Administrator routes at desktop (1440×900), tablet (768×1024), and mobile (390×844), including dashboards, Ticket Detail, Staff Queue, Admin User Management, and Ticket Review. The owner-confirmed checklist records keyboard traversal, visible focus, semantic labels, announcements, non-color status cues, and no clipping, overlap, or horizontal overflow. Axe-core samples reported zero violations and zero incomplete checks on the sampled pages; screenshots alone do not establish authorization.

Screen	Desktop	Tablet	Mobile
Requester Dashboard	1440	768	390
IT Staff Dashboard	1440	768	390
Requester Ticket Detail	1440	768	390
IT Staff Ticket Detail	1440	768	390
Staff Queue	1440	768	390
Admin Dashboard	1440	768	390
Admin User Management	1440	768	390
Admin Ticket Review	1440	768	390

The table links to the full-frame source captures; the following grouped pages compare desktop, tablet, and mobile captures of the primary Lab 4 screens. Action Taken create/edit panels are captured at all three viewport sizes. Full-page images remain available through the VIS-01 links above. The full UI specification, completed visual/accessibility checklist, machine-readable E2E-04 report, and full-resolution visual evidence directory remain available in the repository.

Requester Dashboard – desktop, tablet, and mobile

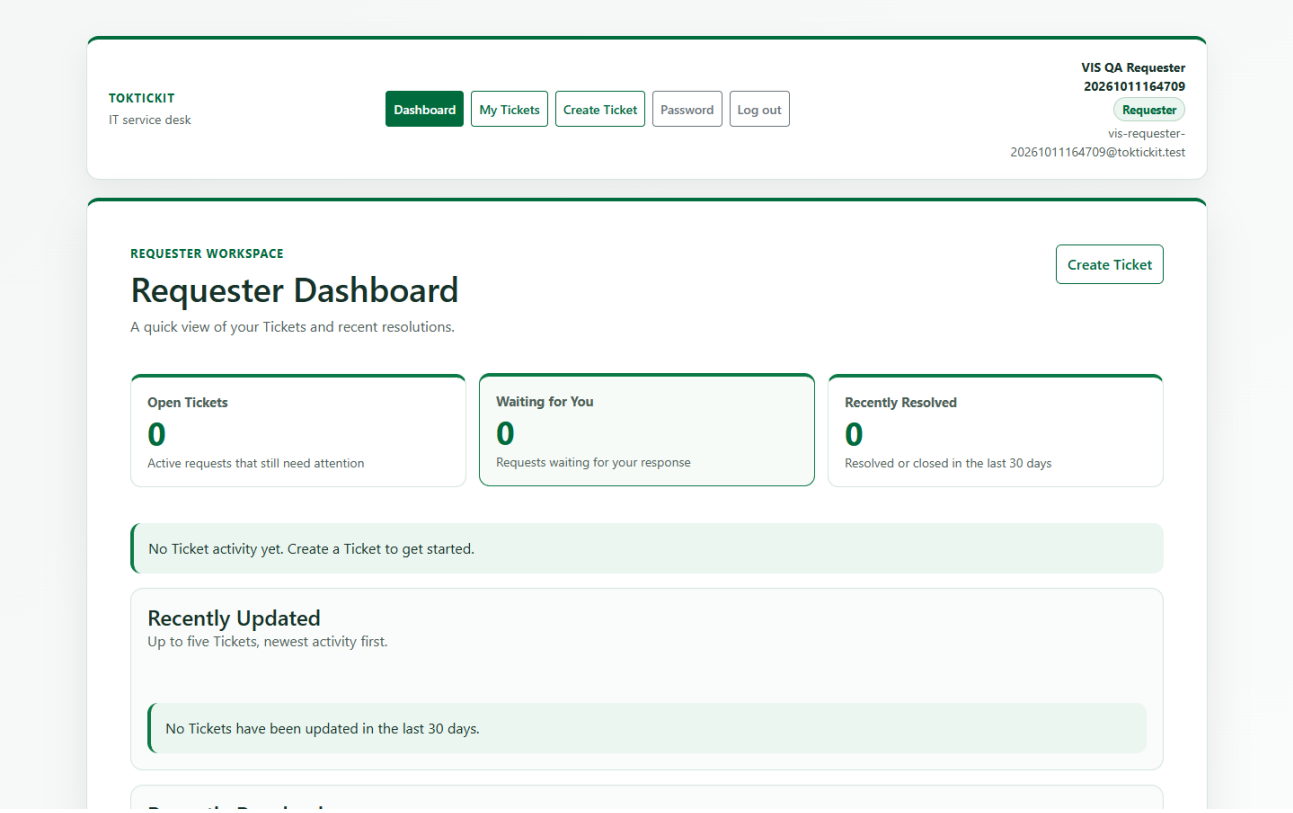


Figure 9: Desktop viewport, 1440 × 900.

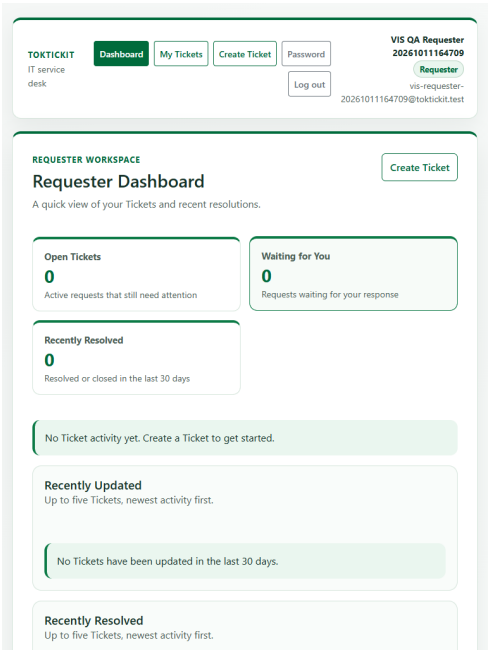


Figure 10: Tablet viewport, 768 × 1024

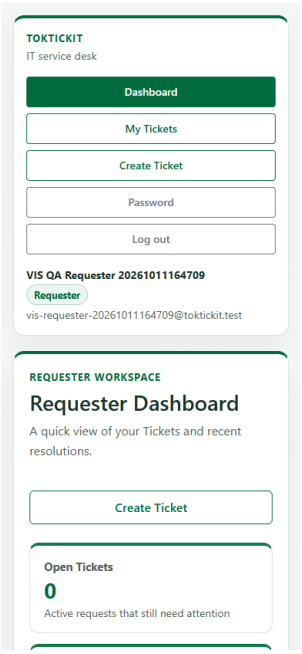


Figure 11: Mobile viewport, 390 × 844

IT Staff Dashboard – desktop, tablet, and mobile

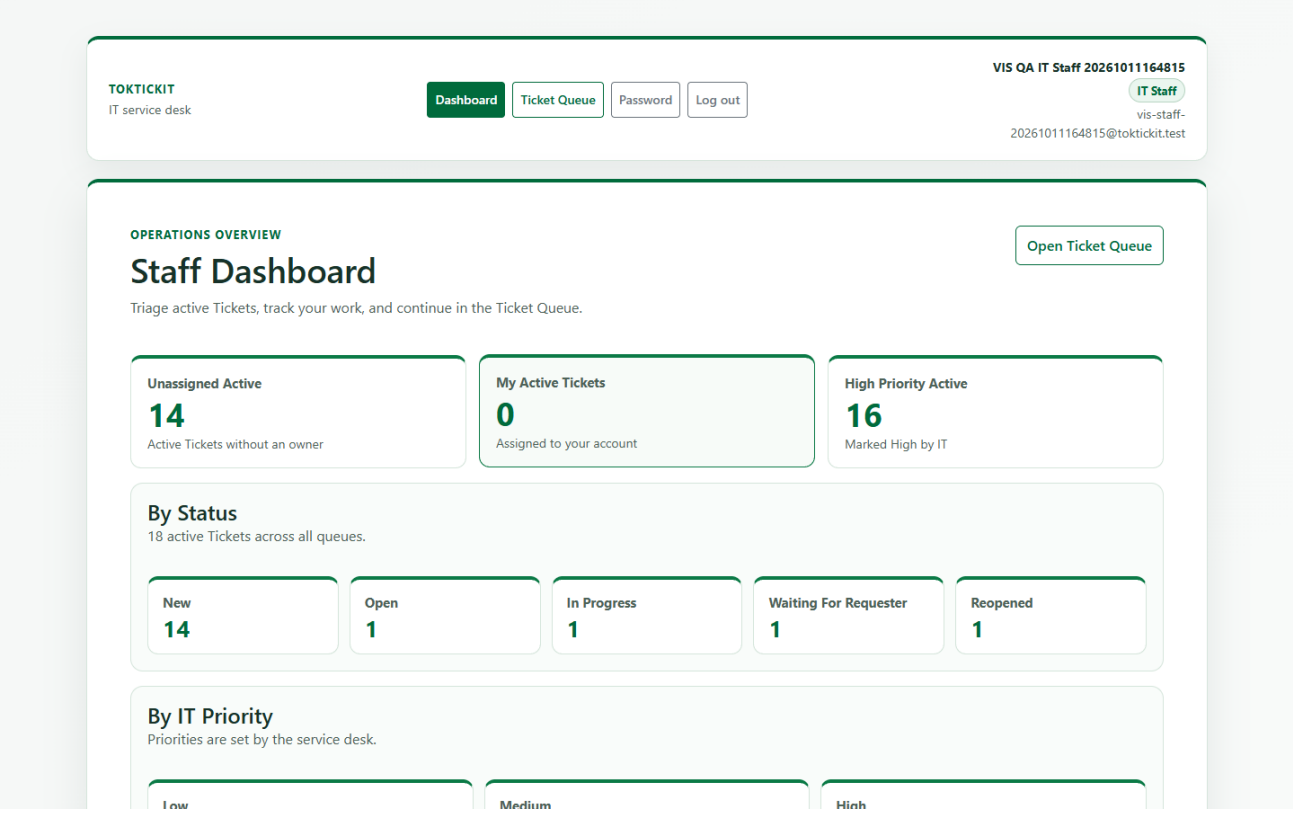


Figure 12: Desktop viewport, 1440 × 900.

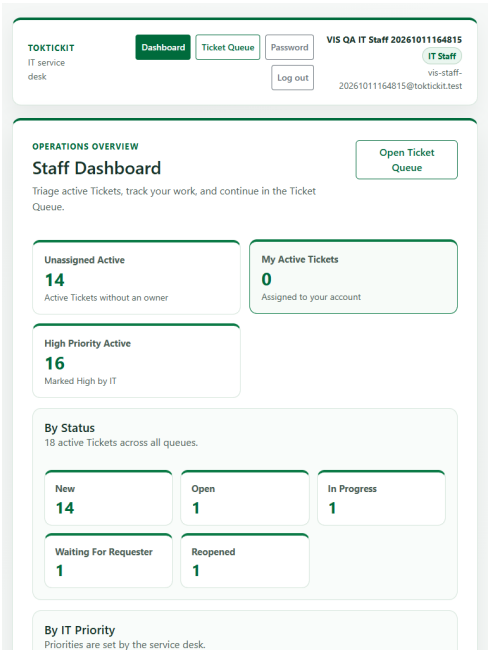


Figure 13: Tablet viewport, 768 × 1024

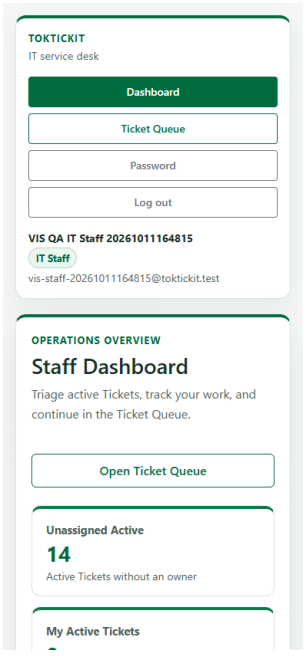


Figure 14: Mobile viewport, 390 × 844

Administrator Dashboard – desktop, tablet, and mobile

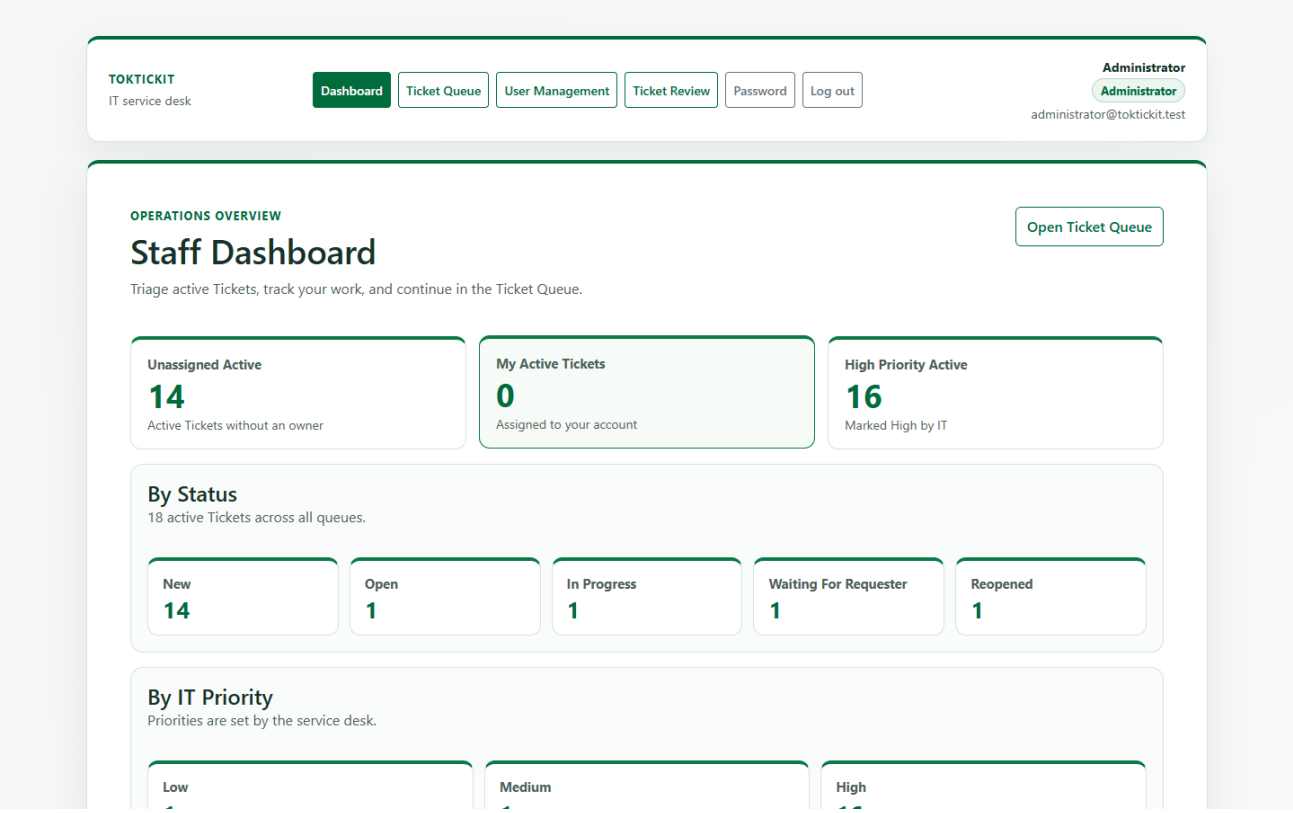


Figure 15: Desktop viewport, 1440 × 900.

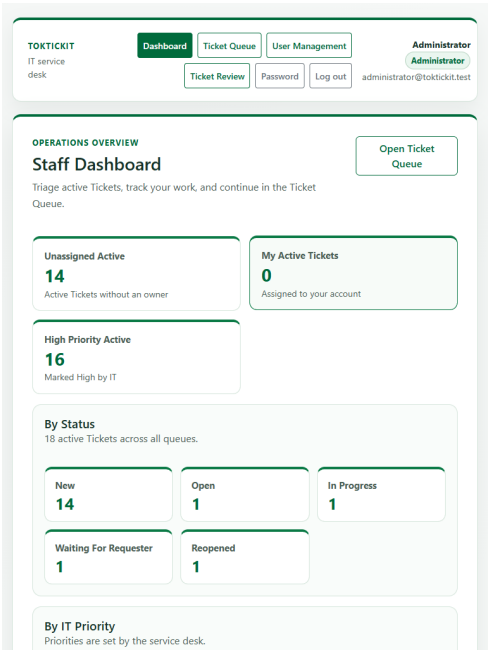


Figure 16: Tablet viewport, 768 × 1024

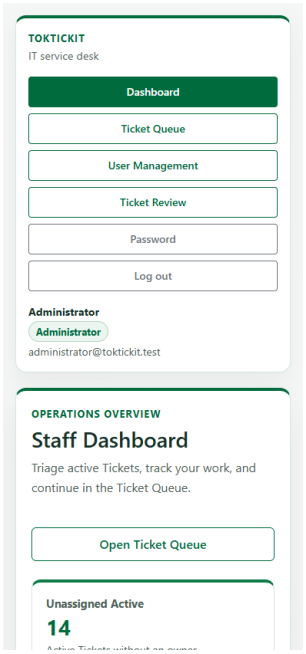


Figure 17: Mobile viewport, 390 × 844

Requester Ticket Detail – desktop, tablet, and mobile

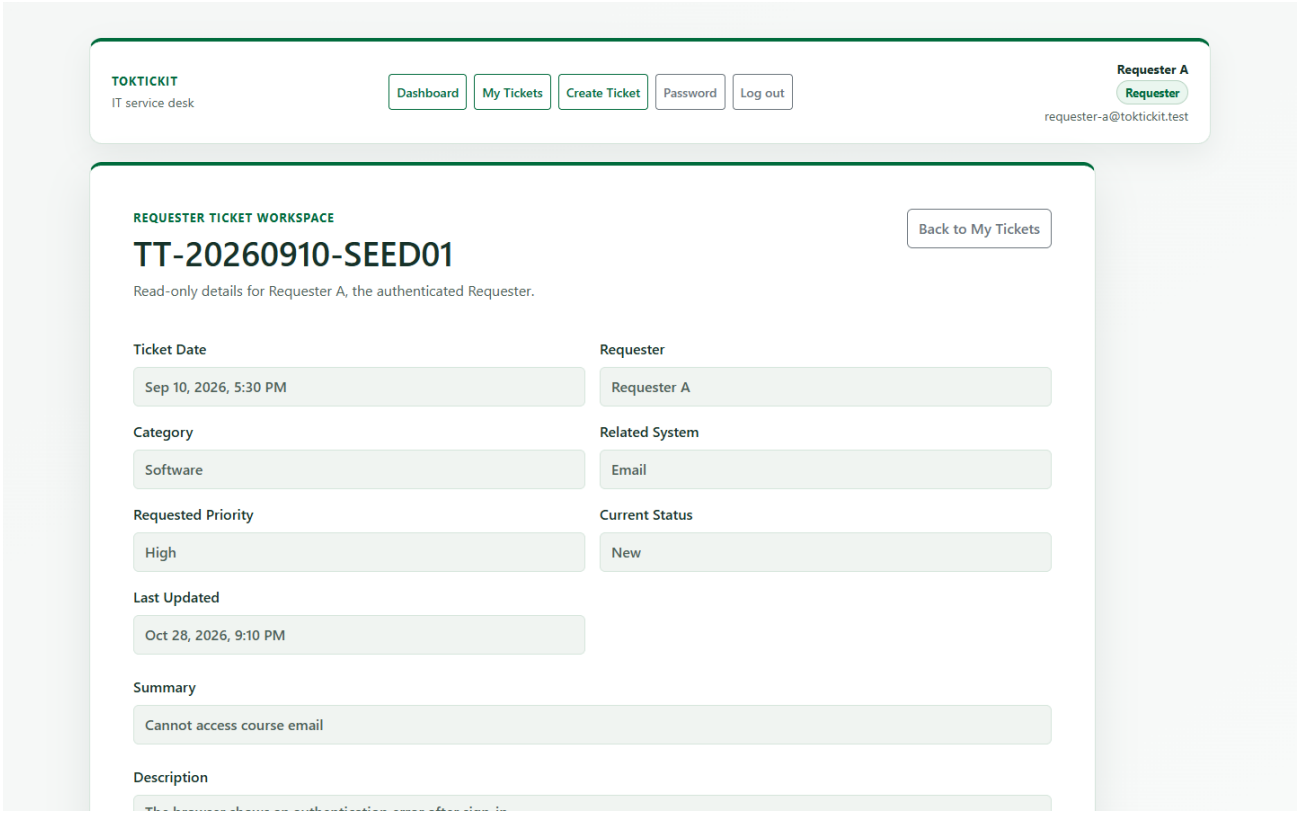


Figure 18: Desktop viewport, 1440 × 900.

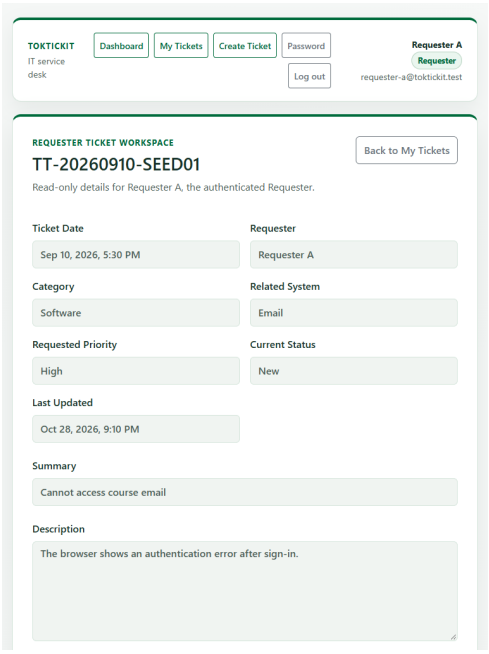


Figure 19: Tablet viewport, 768 × 1024

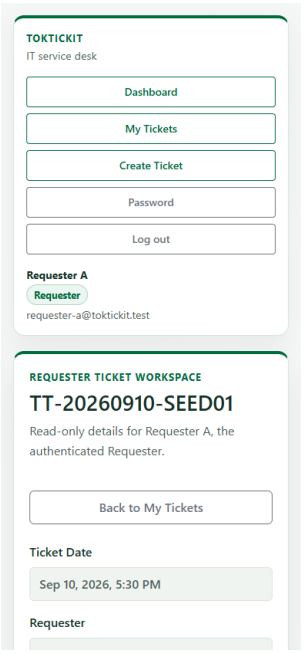


Figure 20: Mobile viewport, 390 × 844

IT Staff Ticket Detail – desktop, tablet, and mobile

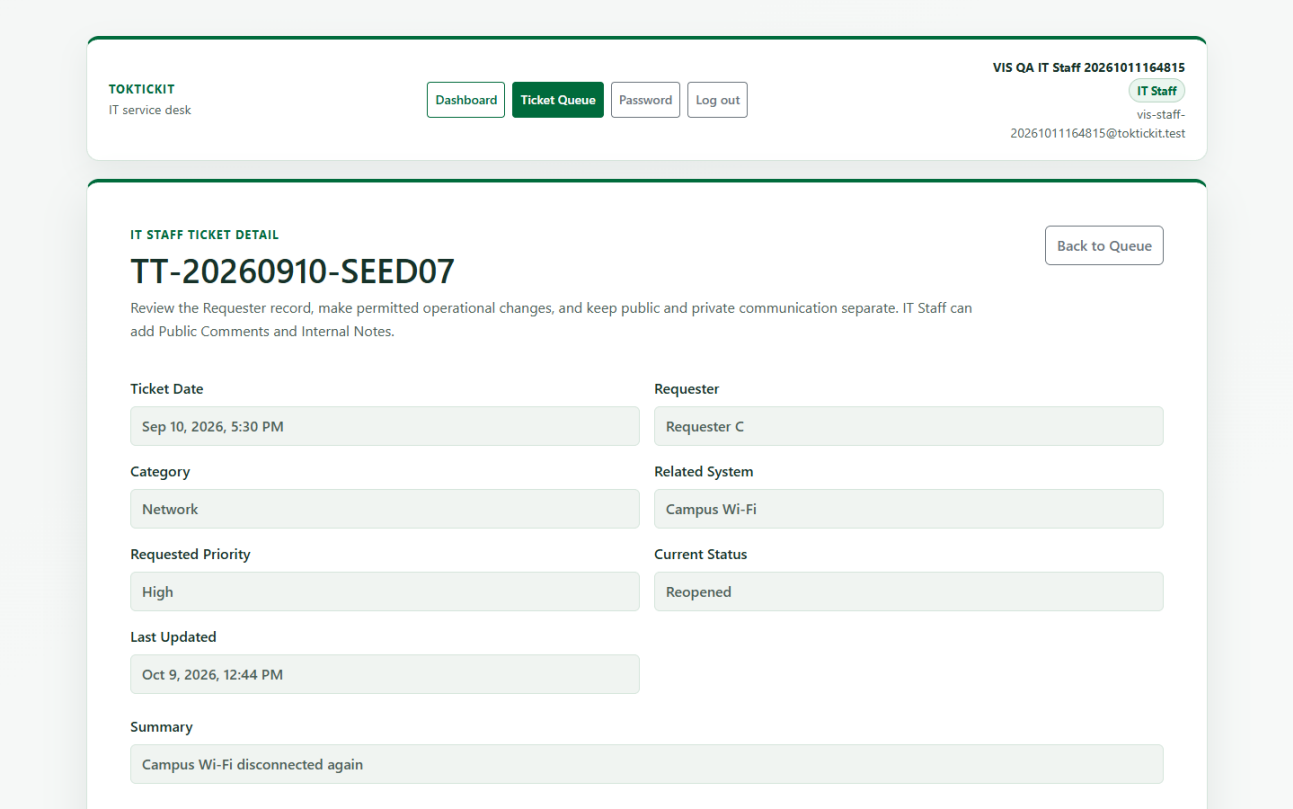


Figure 21: Desktop viewport, 1440 × 900.

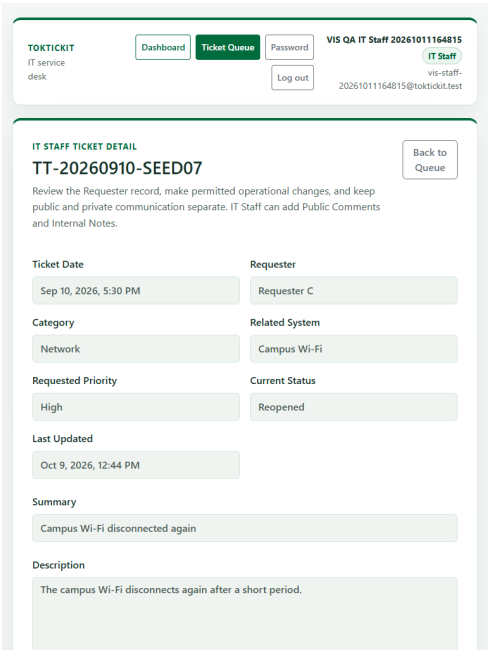


Figure 22: Tablet viewport, 768 × 1024

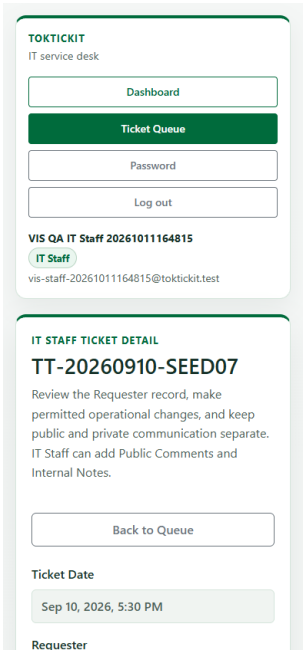


Figure 23: Mobile viewport, 390 × 844

Action Taken create form – desktop form capture

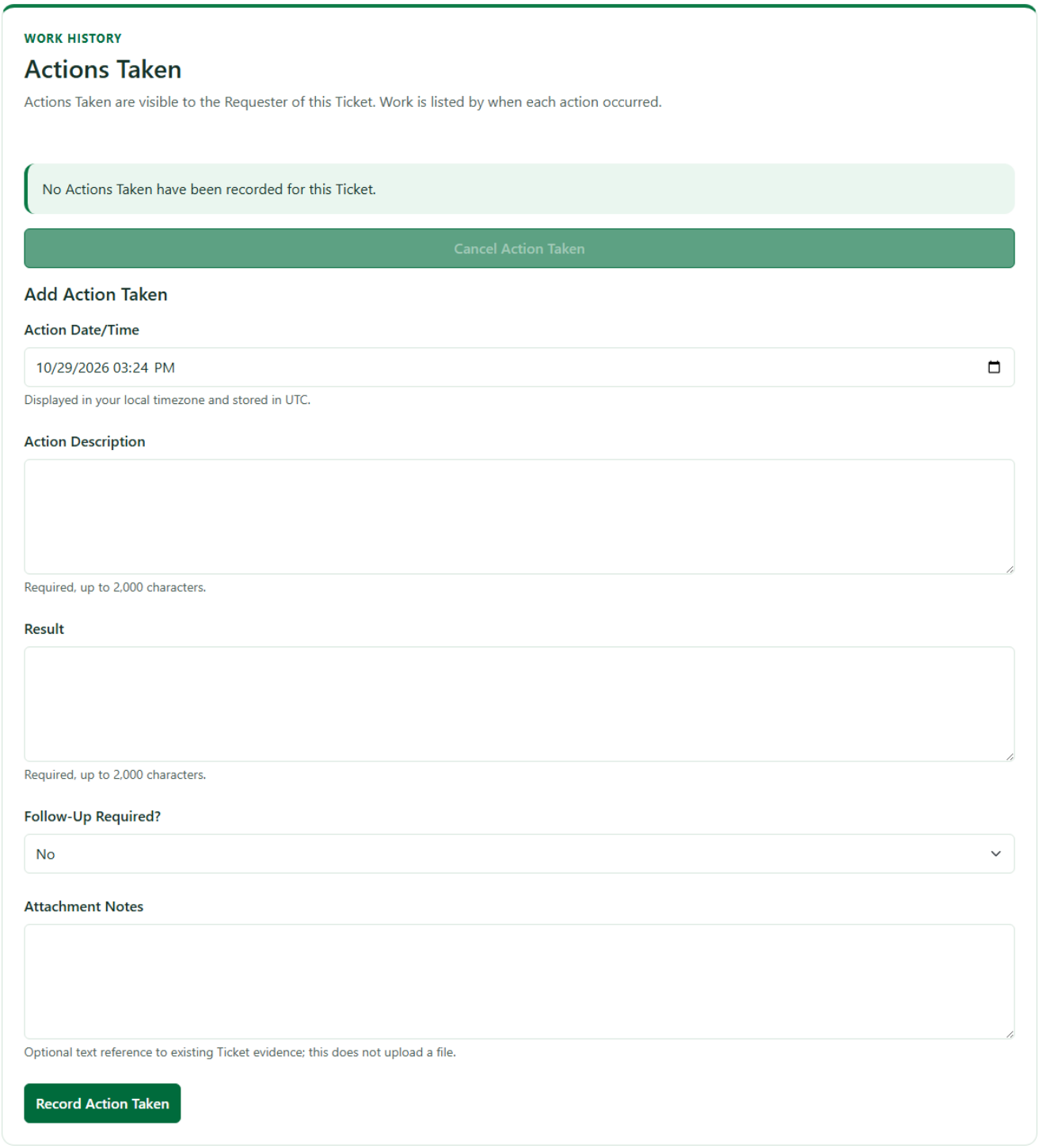


Figure 24: Action Taken panel at 1440 × 900.

Action Taken create form – tablet and mobile form captures

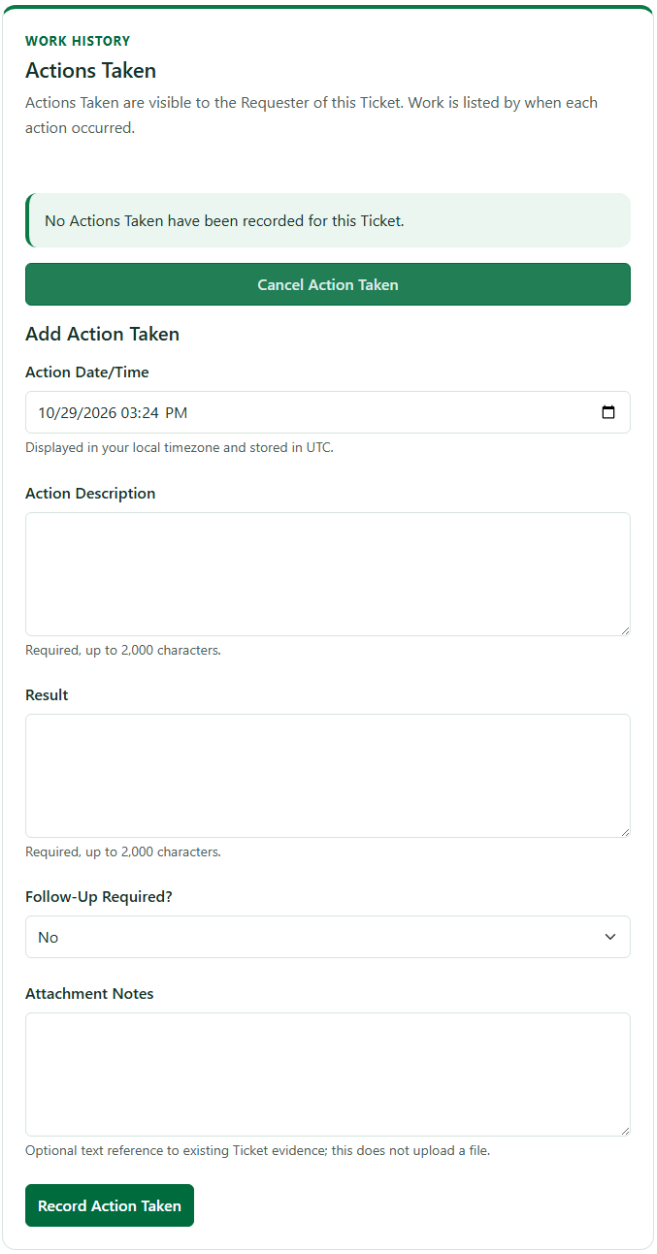


Figure 25: Action Taken panel at 768 × 1024

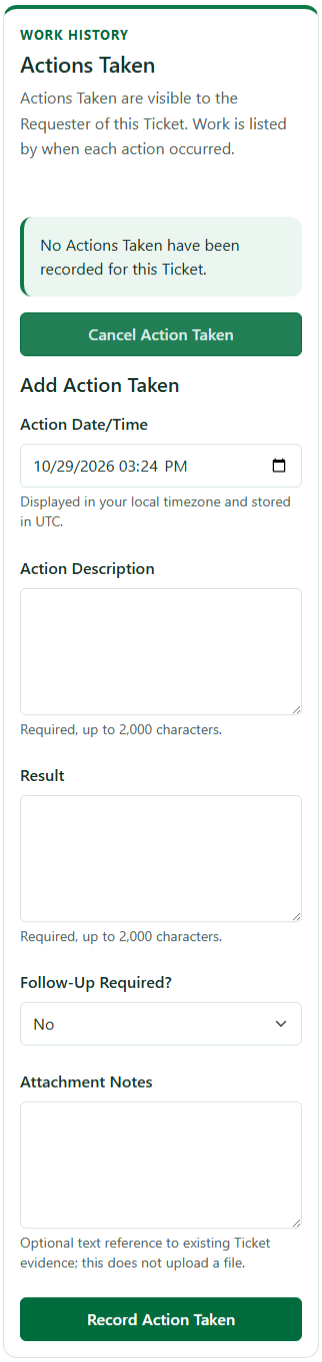


Figure 26: Action Taken panel at 390 × 844

Action Taken edit form – desktop form capture

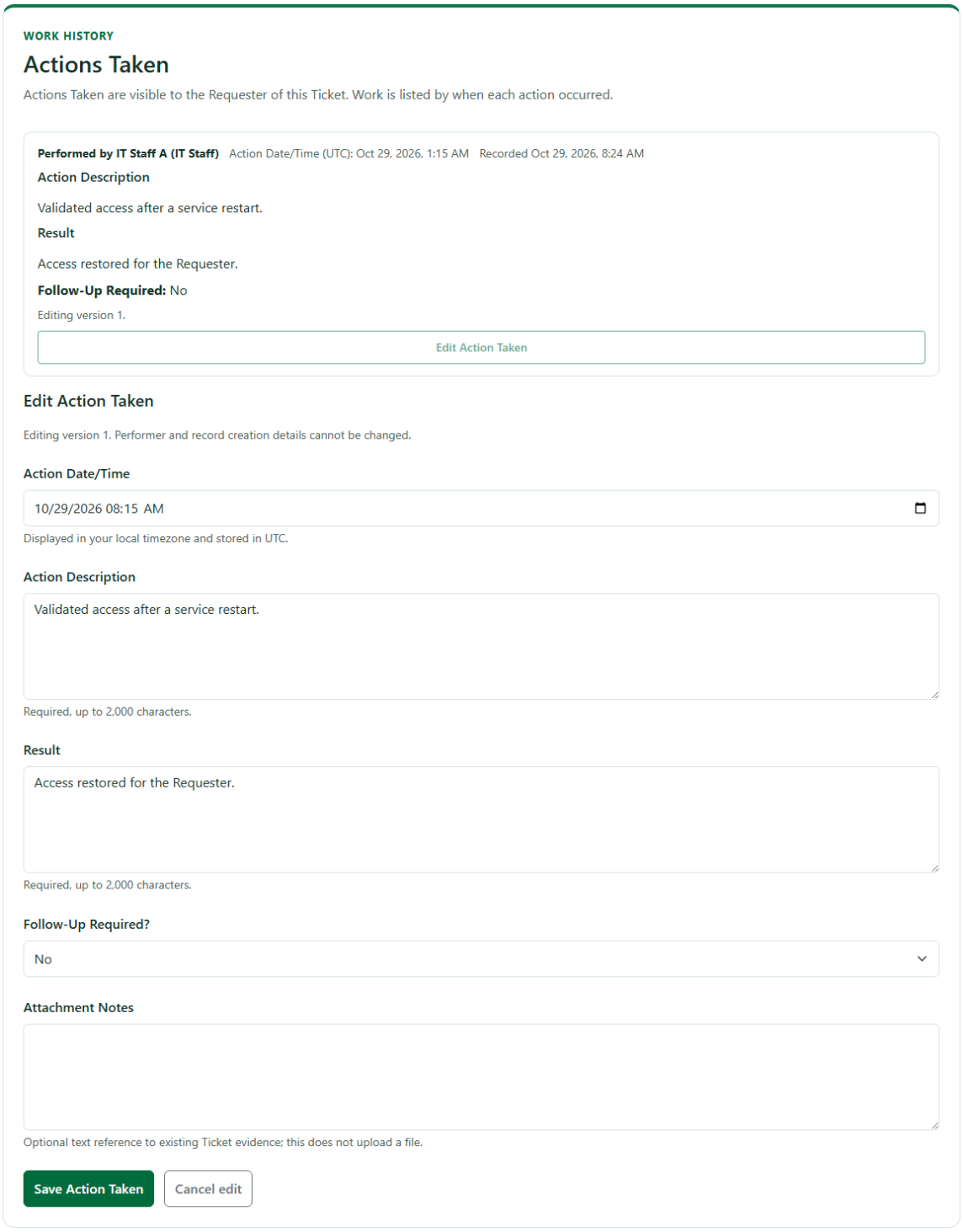


Figure 27: Action Taken panel at 1440 × 900.

Action Taken edit form – tablet and mobile form captures

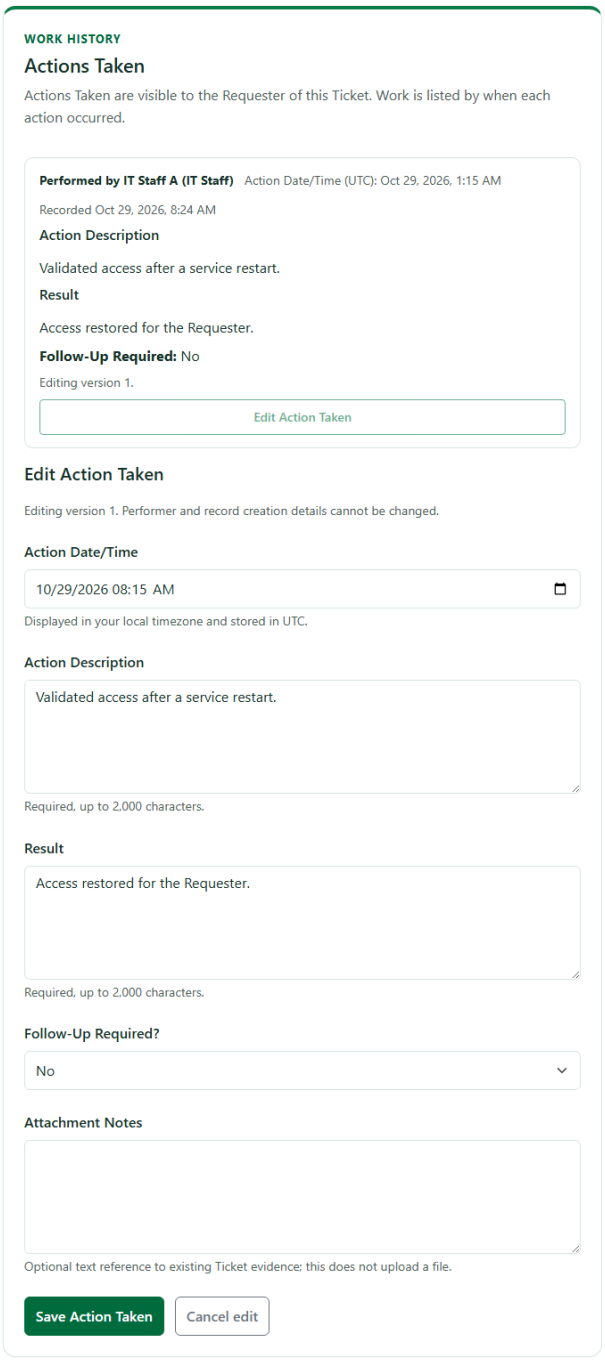


Figure 28: Action Taken panel at 768 × 1024

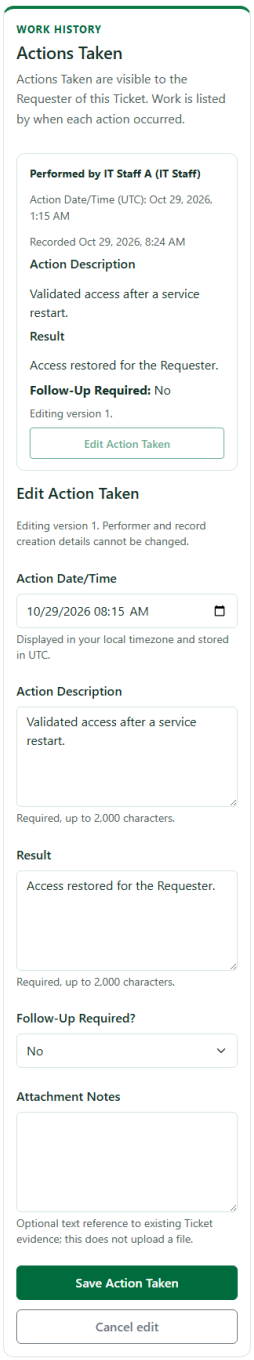


Figure 29: Action Taken panel at 390 × 844